

Find the magnitude and direction angle of the resultant vector.

9) $\vec{a} = \langle 6, -10 \rangle$
 $\vec{g} = \langle -3, 11 \rangle$
Find: $\vec{a} + \vec{g}$

10) Given: $T = (10, -4)$ $X = (0, -1)$
 $Y = (0, -6)$ $Z = (-1, 5)$
Find: $-\vec{TX} - \vec{YZ}$

11) $\vec{u} = \langle 12, 16 \rangle$
Find: $-5\vec{u}$

12) Given: $P = (-4, 3)$ $Q = (6, -9)$
Find: $9\vec{PQ}$

13) $\vec{a} = \langle -3, -12 \rangle$
 $\vec{b} = \langle 4, 9 \rangle$
Find: $-3\vec{a} - 7\vec{b}$

14) Given: $A = (10, 0)$ $B = (3, -1)$
 $C = (-5, 7)$ $D = (-2, 9)$
Find: $-3\vec{AB} + 2\vec{CD}$

Draw a vector diagram to find the resultant of each pair of vectors using the triangle method. Then state the magnitude and direction angle of the resultant.

15) $\vec{m} = \langle -7, 1 \rangle$ $\vec{n} = \langle 12, -16 \rangle$